sru S R U N IVE R S ITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 17.1

Program	: B. Tech (CSE)
Specialization	
Course Title	: Al Assisted coding
Course Code	
Semester	11
Academic Session	: 2025-2026
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Task 1 — Customer Data Cleaning

Task:

Use Al to generate a Python script for cleaning a customer dataset.

Instructions:

- Handle missing values in columns (age, income, city).
- Remove duplicate records.
- Standardize text values (e.g., city names in lowercase). •
  Encode categorical variables (gender, city).

Code:

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
import pandas as pd from
sklearn.preprocessing import
LabelEncoder

# Load dataset data = pd.
read_csv(     "shopping_t
rends. csv" )

print
("Original
Data:" )
print
(data.
head())


      Handle Missing Values (FIXED)


if 'Age' in data. columns:

   datal 'Age'] = data [ 'Age' J. illna(data[   'Age']
.mean())

if 'Purchase Amount (USD) ' in data. columns:
     datal 'Purchase Amount (USD) = data ['Purchase Amount

         (USD) ] . fillna( datal 'Purchase Amount (USD)
         'l.mean()


if 'Location' in data. columns:
     data ['Location'] = data ['Location'] . fillna(data
     ['Location 'J .mode() [0] )

   ------------------------------

# 2. Remove Duplicates
```

```
# ------------------------------
data = data.drop_duplicates()
                    ------------------------------
      Standardize Text

  ------------------------------
if Location' in data. columns:
    data ['Location'] = data ['Location'] .str. lower()

if 'Gender' in data. columns:
    data [ 'Gender'] = data [ 'Gender'] . st r. lower()
```

```
# ------------------------------
# 4.Encode Categorical Variables
# ------------------------------
le = LabelEncoder()

if 'Gender' in data. columns:
    data [ 'Gender' ] = le.fit_transform(data['Gender'] )

if 'Location' in data. columns:
    data [ Location' ] = le. fit transform(data ['Location ' ] )
                                  _
# ------------------------------
# Final Output
# ------------------------------
print("\nCleaned    Data:"    )
print (data. head() )

data. o_csv("cleaned_ custome r_data. csv" , index=Fatse)
```

Output:

urxgxnau vata:

```
   Customer ID Age Gender Item Pu rchased Category Purchase Amount (USD) \1
                          55     Ma le    Blouse Clothing          53
1          2     19     Ma le    Sweater Clothing         64
2          3     50     Ma le    Jeans Clothing           73
3          4     21     Ma le    Sandals Footwear         90
4          5     45     Ma le    Blouse Clothing          49

        Location Size        Color Season Review Rating Subscription Status \
        Kentucky L Gray Winter 3.1 Yes 1 Maine L Ma roon Winter 3.1 Yes
2       Massachusetts  s        Ma roon Spring     3.1        Yes
3       Rhode Island   M        Ma roon Spring     3.5        Yes
4       Oregon         M Turquoise Spring  2.7        Yes

 Payment Method Shipping Type Discount Applied Promo Code Used \
0 Credit Card Express Yes Yes 1 Bank Transfer Express Yes Yes
2          Cash Free Shipping     Yes        Yes
3          PayPal     Next Day Air        Yes        Yes
4          Cash Free Shipping     Yes        Yes

   Previous Purchases Preferred Payment Method Frequency of Purchases
                                        Venmo        Fortnightly 1
  2          Cash       Fortnightly
2                23   Credit Card  Weekly
3                49 PayPal Weekly 4 31 PayPal Annually
```

Cleaned Data:

Customer ID Age Gender Item Purchased Category Purchase Amount (USD) \1

		55	1	Blouse Clothing	53
1	2	19	1	Sweater Clothing	64
2	3	50	1	Jeans Clothing	73
3	4	21	1	Sandals Footwea r	90
4	5	45	1	Blouse Clothing	49

Cleaned Data:

Customer ID Age Gender Item Pu rchased Category Purchase Amount (USI

	1	55	1	Blouse Clothing	53
1	2	19	1	Sweater Clothing	64
2	3	50	1	Jeans Clothing	73
3	4	21	1	Sanda Is Footwear	90
4	5	45	1	Blouse Clothing	49

Location Size Color Season Review Rating Subscription Status 16 Gray Winter 3.1 Yes

1	18	L	Maroon Winter	3.1	Yes
2	20	s	Maroon Spring	3.1	Yes
3	38	M	Maroon Spring	3.5	Yes
4	36	M Turquoise Spring	Yes		

Payment Method Shipping Type Discount Applied Promo Code Used

Credit Card Express Yes Yes I Bank Transfer Express Yes Yes 2

Cash Free Shipping Yes Yes

3	PayPal	Next Day Air	Yes	Yes
4	Cash Free Shipping	Yes	Yes	

Previous Purchases Preferred Payment Method Frequency of Pu rchases

	14	Venmo	Fortnight ly
1	2 Cash	Fortnightly	
2	23 Credit Card	Weekly	
3	49 PayPal	Weekly	
4	31PayPal	Annually	

Task 2 — Sales Data Preprocessing

Task:

Preprocess sales transaction data using Al assistance.

Instructions:

- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset. Code:

```
[5] import pandas as pd import numpy as np from sklearn.
    preprocessing import MinMaxScaler

    data = pd. read_csv("sales_data. csv")
    print ( "Columns : " , data. columns)

    # ------------------------------
    # Auto detect date column

    # ------------------------------
    date_col = None for col
    in data. columns:
        if 'date' in col. lower() :
            date col = col break

    # Convert to datetime data [date coll =
    pd.to_datetime(data [date_col] )

    # Create features data ['year'] = data [date_col]
    .dt.year data [ 'month = data [date_col] .dt.month
    data = data [date_col] .dt.day_name( )

    # ------------------------------
    # Detect sales column

    # ------------------------------
```

```
sales_col = None for cot
in data. columns:
    if 'sale' in col. lower() :
        sales col = cot break


# Normalize scaler = MinMaxScaler( ) data [ ' sales normalized ] = scaler.
it_transform(data[[sales col]])
_
# ------------------------------
# Outlier Removal (IQR)
# ------------------------------
QI = data [sales_col] .quantile(0.25)
Q3 = data [sales_col] .quantile(Ø.75)
IQR = Q3 - QI

lower = QI - 1.5 * IQR upper
= Q3 + 1.5 * IQR

data = data [ (data [sales_col]        lower) & (data [sales_col]        upper) ]

# Output print("\nProcessed
Data:" ) print (data. head( ) )

data. to_csv ( "p rocessed_sa les_data. csv" , index=False)
```

Output:

```
Columns: Index( [ 'Product_        'Sale Date' , 'Sales_Rep' ,        ' Region' , 'Sales Amount ' ,
'Quantity_Sold ' , 'Product_Category' , 'Unit Cost' , 'Unit Price' ,
        ' Customer_Type , 'Discount' , ' Payment_Method ' , 'Sales Channel' ,
        '  Region_and_Sales_Rep  '  ] ,
      dtype= [1] obj ect [I])
Processed Data:
    Product ID Sale Date Sa les_Rep Region     Sales Amount   Quantity_Sold   \
                    _
         1052 2023-02-03          Bob North          5053.97                 18
1        1093 2023-04-21  Bob      West     4384.02  17
2        1015 2023-09-21  David    South     4631.23  30
3        1072 2023-08-24  Bob      South     2167.94  39
4        1061 2023-03-24  Cha rlie East      3750.20  13

   Product_Category    Unit Cost    Unit Price Customer_Type    Discount   \
         Furniture 152.75 267.22 Returning 0.09 1 Furniture
3816.39 4209.44 Returning 0.11
2                Food          261.56   371.40   Returning          0.20
3                Clothing      4330.03 4467.75 New      0.02
4                Electronics          637.37 692.71 New     0.08
  Payment_Method Sales_Channel Region_and_Sales_Rep   year   month day_of_week \
              Cash          Online              North—Bob  2023       2      Friday
1             Cash          Retail              West—Bob   2023       4      Friday
2 Bank Transfer             Retail              South—David 2023       9    Thursday
3       Credit Card         Retail              South—Bob  2023       8    Thursday
4       Credit Card         Online              East—Charlie 2023     3      Friday
    sales normalized o.
090411 1   0.301370
2              o. 720548
3              o. 643836
4              o. 224658
```

Task 3 — Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using Al scripts.

Instructions:

- Handle missing values in blood_pressure, cholesterol.
- Encode categorical features
- Scale numerical features (min-max or standard scaling). •
  Split dataset into training and testing sets.

Code:

```
import pandas as pd from sklearn. preprocessing import
LabelEncoder, MinMaxScaler from sklearn.model_selection
import train_test_split
# Load dataset data = pd.
read_csv("dirty_v3_path.csv")
print ("Original Data:" )
print (data. head() )
_____________________________
# Check columns (IMPORTANT)
_____________________________
print ( "\nCoIumns : " , data. columns)
_____________________________
# 1. Handle Missing Values
_____________________________
# Auto—detect columns for
cot in data. columns:
    if 'blood' in cot. tower() :
        data [cot] = data [cot] . fillna(data [cot] . mean( ) )
    if 'chol' in cot. tower( ) :
        data [cot] = data [cot] . fillna(data [co I] . mean( ) )
# ------------------------------
# 2. Encode Categorical Features
# ------------------------------
le = LabelEncoder()

for cot in data. columns:
    if data [cot] .dtype        'object' :
        data [cot] = le. fit transform(data [cot] )
                                _
# ------------------------------
# 3. Scale Numerical Features (Min—Max)
# ------------------------------
scaler = MinMaxScaler( )

num cols = data.elect_dtypes(include=[' int64 , 'float64']). columns data
[num cols] = scaler.fit_transform(data [num_cols] )
# ------------------------------
# 4. Split Data (Train/ Test)
# ------------------------------
# Assume last column is target (you can change if
needed) X =data. i loc [: , :-IJ y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    , y, test size=ø.2, random state=42

_____________________________
# Final Output
_____________________________
print("\nProcessed    Data:"    )
print (data. head())
```

```
print("\nTrain shape:" , X_train. shape) print
("Test shape:" , X_test. shape)

# Save file data. o_csv("processed_healthcare_ data.csv",
index=False)
```

Output:

```
Original Data:
    Age    Gender Med ica Cond it ion Glucose    B lood P res su re
0 46 ø     Ma I e          Diabetes   137 • ø4          135. 27 28. gø
1 22. ø    Ma I e    Hea hy    71.58    113. 27 26. 29
2 5ø.ø     NaN       Asthma    95 • 24  NaN      22. 53
3 57.ø     NaN       Obes it y          NaN      13ø. 53  38 .47
4 66 . ø Fema le     Hype rtension      95 • 15  178. 17   31. 12
    Oxygen Satu rat ion LengthOfStay    Cho       ro     T r ig lyce r ides HbAI c \
                96. ø4          6          231. 88          21ø . 56    7.61
1               97 . 54         2          165. 57  129 .41  4.91
2               9ø.31           2          214. 94  165 .35  5.6ø
3               96. 6ø          5          197. 71  182. 13  6.92
4               94 . 9ø         4          259. 53  115.85   5.98
    Smoking Alcohol PhysicalAct iv it yDietSco re Family Histo ry \
            ø                          —ø . 2ø        3. 54
            ø
            ø       18. 12                            5. 9ø
            ø
            ø       25. ø1                            4. 65              1
33. 16      3. 37 41 3.56

    St ress Level S teep Hou rs random notes          n o is e_coι
            5.ø7        6.ø5        o rem —137. ø57211
1           5.87        7.72        ips um    —11. 23ø61ø
2                       4. 82       ips um    98.331195
3           3.ø1        5.33        I o rem   44. 187175
4           6.38        6 • 64      I o rem   44. 831426
Columns : Index( [ Age'  'Gender' ,  'Medical Condition' , 'Glucose'  •        Blood
,                                                                          Pressure'        ,
          'BMI ' , ' Oxygen Saturation' , ' LengthOfStay' , 'Cholesterol' ,
          'Triglycerides' , 'HbA1c', ' Smoking '• A Icohol' , • Physical Act ivity' ,
          'Diet Score' , • Family History'        'St  ress  Level'  ,  'Sleep
          Hours , ' random notes '        • noise_col ] ,
Processed Data:
        Age Gender Medical Condition Glucose Blood Pressure BMI 0 o. 455696 0.5
0.428571 o. 391428 o. 401144 o. 431680 1 o. 151899 0.5 0.571429 o. 171904 o. 256540
0.378609
2   o. 506329    1.0        o. 142857 o. 251249  0.435226 0.302155
3   0.594937     1.0        0.857143  NaN        o. 369988 o. 626271
4   o. 708861    ø.ø        o. 714286 o. 250947  o. 683121 0.476820

    Oxygen Saturation LengthOfStay Cholesterol Triglycerides        HbA1c \
            o. 670348      o. 277778      0.518390        o. 524877 0.476872
1           o. 705592      o. 055556  o. 265915  o. 342102 o. 179515
2           o. 535714      o. 055556  0.453891   o. 423050 0.255507
3           o. 683506      o. 222222  o. 388288  0.460844 o. 400881
4           o. 643562      o. 166667  o. 623667  0.311561 o. 297357

    Smoking Alcohol Physical Activity Diet Score Family History \
        0.0        ø.ø          0.216283      o. 383056              ø.ø
1                               0.733375      o. 553946  ø.ø
2                               0.0 ø.ø       o. 540087  o. 463432  1.0
3                               0.425109      0.370746   ø.ø
4                               0.0 1.0       0.449969   0.372918   ø.ø
```

	Stress Level	Sleep Hours	random_notes	noise col
	0.419788	o. 509132	1. øøøøøø	o. 312605
1	0.464505	o. 699772	o. 666667	0.455579
2	0.309111	o. 368721	o. 666667	o. 580072
3	0.304639	0.426941	1. øøøøøø	0.518549
4	0.493013	o. 576484	1. øøøøøø	0.519281

Train shape: (24000, 19)
Test shape: (600, 19)

Task 4 — Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using Al scripts.

Instructions:

• Handle missing values

• Detect and remove salary outliers.

• Standardize department names (lowercase).

• Apply label encoding to job location, department.

Code:

```
import pandas as pd from sklearn. preprocessing
import LabelEncoder

# Load dataset data = pd.
read_csv("Employee_Salary_Dataset. csv" )

print ("Original Data: ")
print (data. head( ) )

# Check columns (IMPORTANT)

print ("\nColumns : , data. columns)

# ------------------------------
# 1. Handle Missing Values

for cot in data. columns:
    if data [cot] .dtype in int64% 'float64'] :
        data [cot] = data [cot] . fillna(data [co l] . mean() ) else:
        data [cot] = data [cot] . fillna(data[col] . mode( ) [0] )

# ------------------------------
# 2. Detect & Remove Salary Outliers (IQR)

# Auto detect salary column
salary_col = None for col in
data. columns: if 'salary' in col.
lower() : sa la ry_col = col
break

QI = data [salary_col] . quantile(Ø.25)
Q3 = data [salary_cot] . quantile(0.75)
IQR = Q3 - QI

lower = QI - 1.5 * IQR upper =
03 + 1.5 * IQR

data = data [ (data [salary_col] lower) & (data [salary_col] <= upper) ]

# ------------------------------
# 3. Standardize Department Names

# ------------------------------
for col in data. columns:
    if 'department' in col. lower() :
        data [cot] = data [cot] . str. lower( )
```

```
# ------------------------------
# 4.Label Encoding
# ------------------------------
le = LabelEncoder()
for col in data. columns:
    if data [cot] .dtype        'object' :
        data [cot] = le. fit_t ransform (data [cot] )
```

```
# Final Output
# ------------------------------
print("\nProcessed    Data:"    )
print (data. head())

# Save file data. to_csv ( processed_employee_data.csv" ,
index=Fatse)
```

Ouput:

Original Data:

	ID	Experience_Years	Age	Gender	Sa la ry
ø	1	5	28	Female	25øøøø
1	2	1	21	Male	5øøøø
2	3	3	23	Female	17øøøø
3	4	2	22	Male	25000
4	5	1	17	Male	løøøø

Columns: Index(['ID', 'Experience_Years' ,'Age' , 'Gender' , 'Salary'], dtype='object')
Processed Data:

	ID	Experience_Years	Age	Gender	Sa ry	la
0	1	5	28	o	250000	
1	2	1	21			
2	3	3	23	17øøøø		
3	4	2	22	1	25000	
4	5	1	17	1	løøøø	